Giga-Embeddings: Mixture-of-Experts Encoders for High-Throughput Text Embeddings

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Abstract

We introduce Giga-Embeddings, a family of text embedding models designed to combine strong retrieval quality with efficient serving. Its largest member is a sparse 10B-parameter Mixture-of-Experts encoder that activates a subset of its parameters for each token. Across English, Russian, multilingual, and code MTEB benchmarks, the MoE model achieves the strongest aggregate performance within the family on all four evaluated suites. In our vLLM benchmark, its throughput is 28%, 25%, and 13% higher than that of the dense 3B model at input lengths of 512, 1024, and 2048 tokens, respectively. At 1024 tokens it reaches 114.5k tokens per second, 1.56–2.65× the throughput of the evaluated external systems. The compact 480M model achieves a Russian MTEB score comparable to FRIDA (70.96 vs. 70.95) with approximately 42% fewer parameters and also supports multilingual embedding tasks. The family offers models with different combinations of embedding quality, throughput, and model size.

1 Introduction

Text embeddings support semantic search, retrieval-augmented generation, clustering, classification, and semantic textual similarity. MTEB (Muennighoff et al., 2023) and its multilingual expansion MMTEB (Enevoldsen et al., 2025) provide broad evaluation suites for these use cases. Recent embedding systems adapt decoder-only language models into bidirectional encoders and train them with large-scale contrastive objectives (BehnamGhader et al., 2024; Wang et al., 2024; Lee et al., 2025; Kolodin and Ianina, 2025). Although scaling the backbone generally improves representation quality, billion-parameter encoders are costly to store and serve.

Knowledge distillation offers one route to smaller encoders. For embedding models, supervision can be transferred by matching embeddings or by matching the teacher's relative similarity scores over candidate sets (Reimers and Gurevych, 2020; Xu et al., 2023; Zhao et al., 2026). Matching similarity scores allows supervision to transfer between models with different embedding dimensions and architectures, including dense students and sparse Mixture-of-Experts (MoE) teachers.

We study a family of embedding systems that combines a high-capacity sparse encoder with smaller dense encoders, including a student trained with similarity-distribution distillation. The 10B MoE model uses an estimated 1.63B active parameters per token under the counting convention in Appendix A and exceeds the dense 3B model's measured batch-16 throughput. The distilled 480M model achieves Russian benchmark performance comparable to FRIDA with substantially fewer parameters. It also supports multilingual embedding tasks. Our contributions are:

1. an openly released model family spanning dense 480M and 3B encoders and a sparse 10B encoder;

2. an evaluation on four MTEB suites in which the 10B model has the highest aggregate scores within the family and the highest Russian score among the compared baselines;

3. a compact encoder with a Russian MTEB score comparable to FRIDA (70.96 vs. 70.95) and approximately 42% fewer parameters (480M vs. 823M), evaluated on English, multilingual, and code tasks as well;

4. a throughput comparison of five serving configurations in which the MoE model exceeds dense 3B throughput by 13–28% across three input lengths.

Direct checkpoint links are withheld during anonymous review and will be restored in the camera-ready version.

Table 1: Stage-specific training settings. Variations in the 480M development runs are described in Appendix B. Fine-tuning and multitask batches contain 8,192 candidate texts.

Setting	Pre-training	Fine-tuning	Multitask
Global batch size	16,384	1,024	1,024
Candidates/sample	in-batch	1 pos. + 7 neg.	1 pos. + 7 neg.
In-batch negatives	yes	no	no

2 Related Work

General-purpose text embeddings. Sentence-BERT (Reimers and Gurevych, 2019) and SimCSE (Gao et al., 2021) established bi-encoder and contrastive recipes, subsequently scaled by E5, GTE, and BGE (Wang et al., 2022; Li et al., 2023; Xiao et al., 2023). Instruction-conditioned systems such as INSTRUCTOR, E5-Mistral, and NV-Embed (Su et al., 2023; Wang et al., 2024; Lee et al., 2025) adapt language models to embedding tasks. Qwen3-Embedding (Zhang et al., 2025) uses the same architecture family as our dense encoders.

Embedding distillation. Classical distillation matches output distributions (Hinton et al., 2015). Representation transfer and contrastive self-distillation have been studied for sentence embeddings (Reimers and Gurevych, 2020; Xu et al., 2023), while retrieval methods transfer margins or listwise similarities (Hofstätter et al., 2020; Ren et al., 2021; Lee et al., 2024). KaLM-Embedding-V2 and jina-embeddings-v5-text combine contrastive supervision with distillation (Zhao et al., 2026; Akram et al., 2026). Our 480M student likewise uses KL divergence to match teacher and student similarity distributions.

MoE embedding models. MoE models increase total capacity while activating a subset of experts per token (Shazeer et al., 2017; Fedus et al., 2022; Jiang et al., 2024). Nomic Embed v2 (Nussbaum and Duderstadt, 2025) is a previously released general-purpose MoE text embedding model. MoEE (Li and Zhou, 2025) constructs embeddings from routing weights and hidden states of existing MoE language models without additional fine-tuning. Our study evaluates a trained bidirectional MoE encoder alongside dense models in the same released family.

3 Model and Training Recipe

3.1 Encoder adaptation

Following the earlier Giga-Embeddings recipe (Kolodin and Ianina, 2025), we adapt decoder-only language models to produce text embeddings. All three encoders are initialized from proprietary GigaChat pre-training checkpoints. Qwen3 and DeepSeek refer to the network architectures, not to public initialization checkpoints. The causal attention mask is replaced by a fully visible bidirectional mask, as in LLM2Vec (BehnamGhader et al., 2024), so that each token can use both left and right context. We use mean pooling to aggregate token states into a fixed-dimensional vector and normalize the resulting vector before cosine-similarity scoring. The instruction templates follow the earlier Giga-Embeddings recipe (Kolodin and Ianina, 2025) and distinguish asymmetric retrieval inputs from symmetric semantic-similarity inputs. The 480M, 3B, and 10B-A1.8B checkpoints and instruction set are publicly available; direct repository links are withheld during anonymous review. The released model cards specify the MIT license for all three checkpoints.

The 480M, 3B, and 10B-A1.8B encoders produce embeddings with dimensions 1024, 2048, and 1536, respectively. The 480M and 3B models use dense bidirectional encoders in the Qwen3 architecture family (Zhang et al., 2025), while the largest model is a bidirectional DeepSeekMoE-style encoder (Dai et al., 2024) with 10B total parameters, 64 routed experts, one shared expert, and top-4 routing. The dense encoders have 28 and 36 layers with hidden sizes 1024 and 2048; the MoE encoder has 26 layers with hidden size 1536, sigmoid router scores, and 1280-dimensional expert feed-forward layers. The largest model is released as "10B-A1.8B". Using the counting convention in Appendix A, we estimate 1.63B active parameters per token. We refer to the models as "480M", "3B", and "10B-A1.8B".

3.2 Three-stage contrastive training

As in the first Giga-Embeddings release (Kolodin and Ianina, 2025), training has three stages: broad contrastive pre-training, retrieval fine-tuning with hard negatives, and multitask fine-tuning for retrieval, classification, clustering, and semantic similarity. Embedding training uses open-source data from the Nemotron (Babakhin et al., 2025) and

KaLM (Zhao et al., 2026) recipes. Appendix B identifies the corresponding public source collections and their published roles. The backbone pre-training is proprietary; the datasets used for embedding training are publicly available. Across all stages, the learning rate starts at 3×10^{-5} and follows a cosine decay schedule with a 1% warmup fraction and a minimum learning rate of 10^{-7}. We use weight decay of 0.01, Adam coefficients $\beta_1 = 0.9$ and $\beta_2 = 0.999$, and an InfoNCE temperature of 0.02. Fine-tuning runs for one epoch and multitask training for three epochs. The stage-specific settings are summarized in Table 1. Some 480M development runs use different settings, as detailed in Appendix B.

Our contrastive-loss formulation follows Qwen3-Embedding (Zhang et al., 2025). Pre-training uses in-batch InfoNCE, includes query–query and document–document contrastive terms, and masks detected false negatives. Fine-tuning and multitask training disable in-batch negatives and use the explicit eight-candidate groups described in Table 1. For a batch of B queries, let $\mathcal{C}_i$ be the fixed candidate set for query q_i, let $p(i)$ index its positive candidate, and let s_{ij} be the cosine similarity between encoder outputs for q_i and candidate j.

We use batch-mean InfoNCE (van den Oord et al., 2018):

$$\mathcal{L}_{\mathrm{NCE}} = -\frac{1}{B}\sum_{i=1}^{B} \log \frac{\exp(s_{i,p(i)}/\tau)}{\sum_{j\in\mathcal{C}_i}\exp(s_{ij}/\tau)}. \quad (1)$$

Hard negatives for retrieval fine-tuning are mined from a predefined retrieval-rank range.

3.3 Similarity-distribution distillation

We apply similarity-distribution distillation exclusively to the 480M model during retrieval fine-tuning and multitask fine-tuning. The teacher is a 10B MoE encoder initialized from a proprietary GigaChat pre-training checkpoint. We precompute and cache the teacher's query and candidate embeddings, then use them to calculate similarity targets during student training. The 480M student uses only the contrastive objective during pre-training. The 3B and 10B-A1.8B models are trained without distillation throughout. For both distilled stages, $\lambda = 0.3$, $\tau_{\mathrm{KD}} = 0.05$, and the contrastive temperature is 0.02. Because the objective aligns distributions over query–candidate similarity scores rather than embedding coordinates, the teacher and student may use different embedding dimensions. By contrast, direct MSE or cosine-similarity alignment between teacher and student embedding vectors requires equal dimensions or an additional learned projection. Let t_{ij} be the teacher cosine similarity for the same support $\mathcal{C}_i$. Teacher and student distributions are

$$P_T(j \mid q_i) = \frac{\exp(t_{ij}/\tau_{\mathrm{KD}})}{\sum_{k\in\mathcal{C}_i}\exp(t_{ik}/\tau_{\mathrm{KD}})}, \quad (2)$$

$$P_S(j \mid q_i) = \frac{\exp(s_{ij}/\tau_{\mathrm{KD}})}{\sum_{k\in\mathcal{C}_i}\exp(s_{ik}/\tau_{\mathrm{KD}})}. \quad (3)$$

The distillation term is a batch mean over identical supports:

$$\mathcal{L}_{\mathrm{KD}} = \frac{1}{B}\sum_{i=1}^{B} D_{\mathrm{KL}}(P_T(\cdot \mid q_i) \,\|\, P_S(\cdot \mid q_i)). \quad (4)$$

The combined objective is

$$\mathcal{L} = \lambda\mathcal{L}_{\mathrm{KD}} + (1-\lambda)\mathcal{L}_{\mathrm{NCE}}. \quad (5)$$

The KL term transfers the teacher's relative similarity structure over each candidate set, while InfoNCE preserves direct contrastive supervision. Because the loss depends only on final similarity scores, it does not require access to the teacher's internal architecture or routing decisions.

4 Evaluation

4.1 Evaluation protocol

Unless otherwise noted, scores for our released models are the official MTEB Leaderboard's `meanTask` values retrieved on 2026-10-29. The four suites contain 41 English, 23 Russian, 131 multilingual, and 12 code tasks; each task receives equal weight in the suite-level mean. The Russian suite follows ruMTEB (Snegirev et al., 2025), while the multilingual suite follows MMTEB (Enevoldsen et al., 2025). The leaderboard does not provide a complete multilingual aggregate for 480M. For this model, we recompute the multilingual task-macro mean from stored task-level results, obtaining 56.80†. These results were produced with different MTEB versions and cannot be verified against a specific model revision (Appendix E). The distillation comparison uses the evaluation results of the two training variants. For Russian, that comparison selects the test split when available; this gives 70.98 for the distilled 480M model, compared with the leaderboard's 70.96 when validation and test are averaged for the two MASSIVE tasks (Appendix D).

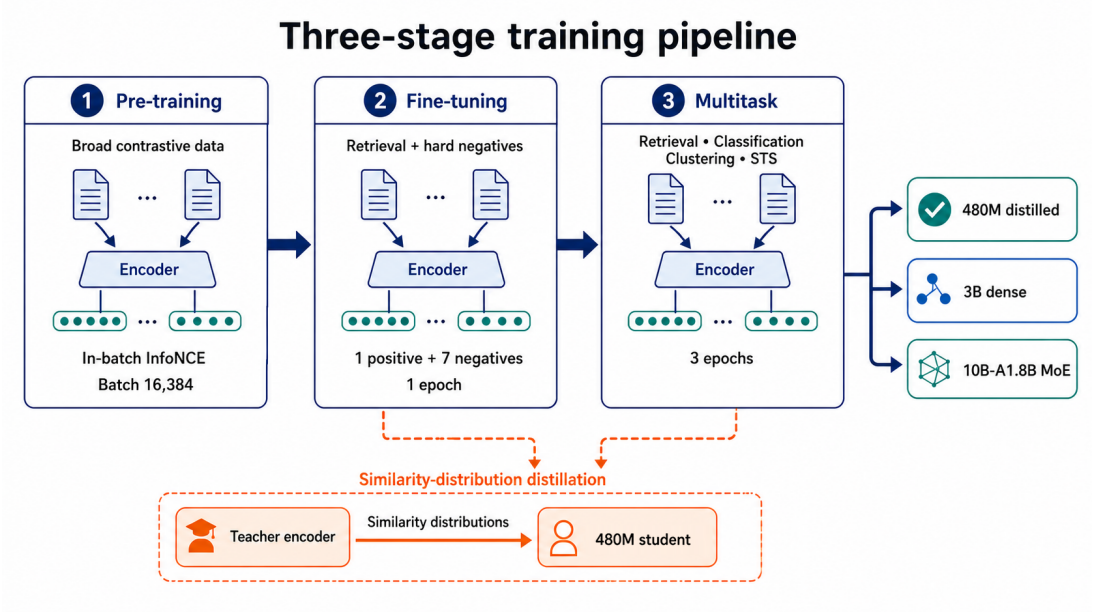


Figure 1: Core three-stage embedding training pipeline. All models undergo contrastive pre-training, retrieval fine-tuning, and multitask fine-tuning. During the latter two stages, only the 480M student additionally matches similarity distributions from a 10B MoE teacher.

Table 2: Model architectures; active counts follow Appendix A.

Type	Total	Active	Dim.
Dense	480M	480M	1024
Dense	3B	3B	2048
MoE	10B	∼1.63B	1536

Table 3: MTEB task-macro means (%). Leaderboard snapshot: 2026-10-29. †: recomputed from stored task results (Appendix E). Bold: highest score within the family.

Scale	Dim.	English	Russian	Multilingual	Code
480M	1024	69.52	70.96	56.80†	72.87
3B	2048	71.90	74.55	63.89	76.94
10B-A1.8B	1536	**72.23**	**74.99**	**65.64**	**78.43**

Table 4: Offline token-ID throughput (tokens/s): one H100 80 GB per run, vLLM 0.24.0, BF16, batch 16. Seven timed repeats after three warm-ups; relative: MoE = 1 at 1024 tokens.

Model (vLLM)	512 tokens	1024 tokens	2048 tokens	Relative
Qwen3-Embedding-4B	70.1k	73.2k	71.2k	0.64×
F2LLM-v2-8B	43.2k	43.4k	42.6k	0.38×
nvidia/llama-embed-nemotron-8b	42.6k	43.2k	41.7k	0.38×
Ours: 3B	87.9k	91.5k	90.4k	0.80×
Ours: 10B-A1.8B	**112.6k**	**114.5k**	**102.3k**	**1.00×**

Table 5: Comparison with compact models on MTEB suites (%). Baselines: official leaderboard `meanTask` values retrieved on 2026-08-08. Our scores are from Table 3. A dash denotes an unavailable score.

Model	Params	Dim.	Russian	English	Multilingual	Code
sergeyzh/BERTA	128M	768	69.39	—	—	—
google/embeddinggemma-300m	308M	768	65.19	69.67	61.15	68.76
codefuse-ai/F2LLM-v2-0.6B	596M	1024	65.97	69.97	62.74	77.41
Qwen/Qwen3-Embedding-0.6B	596M	1024	64.29	70.47	64.34	75.42
ai-forever/FRIDA	823M	1536	70.95	—	—	—
Ours: 480M	480M	1024	70.96	69.52	56.80†	72.87

At two bytes per weight (BF16 or FP16), the 480M, 3B, and 10B-A1.8B models require approximately 0.97, 6.30, and 20.95 GB of weight storage, respectively (decimal GB). The distilled model

Table 6: Comparison with larger models on MTEB suites (%). Baselines: official leaderboard `meanTask` values retrieved on 2026-08-08. Our scores are from Table 3.

Model	Params	Dim.	Russian	English	Multilingual	Code
codefuse-ai/F2LLM-v2-1.7B	1.72B	2048	68.52	71.63	65.21	78.76
codefuse-ai/F2LLM-v2-4B	4.02B	2560	69.46	72.41	67.06	80.15
Qwen/Qwen3-Embedding-4B	4.02B	2560	70.13	74.61	69.45	80.07
Qwen/Qwen3-Embedding-8B	7.57B	4096	71.42	75.23	70.58	80.69
codefuse-ai/F2LLM-v2-8B	7.57B	4096	70.57	72.86	68.09	80.16
codefuse-ai/F2LLM-v2-14B	13.99B	5120	70.90	73.08	68.74	80.75
Ours: 3B	3B	2048	74.55	71.90	63.89	76.94
Ours: 10B-A1.8B	10B	1536	74.99	72.23	65.64	78.43

Table 7: Scores for two 480M training variants (%), with and without distillation. Each row represents one training run.

Model	English	Russian	Code
480M, no distillation	69.43	70.86	72.65
480M, distilled	69.52	70.98	72.87

therefore uses about 4.6% as much weight storage as the MoE model. These theoretical estimates exclude activations and other runtime state.

4.2 Aggregate results

Table 3 presents task-macro scores for all three models. The 480M distilled model scores 69.52 on English, 70.96 on Russian, and 72.87 on code tasks in the leaderboard snapshot. Its recomputed multilingual task-macro mean is 56.80. Using unrounded scores, the English, Russian, and code gaps to 3B are 2.39, 3.59, and 4.07 points. Using the recomputed 480M score, the multilingual gap is approximately 7.09 points.

The 10B-A1.8B model achieves the highest task-macro mean within the family on all four suites. Using unrounded scores, the differences from 3B are +0.33 on English, +0.43 on Russian, +1.75 on multilingual tasks, and +1.49 on code tasks.

4.3 Descriptive distillation comparison

Table 7 compares two 480M training variants using the same test-preferred Russian aggregation. The English, Russian, and code differences are +0.09, +0.12, and +0.22 points, respectively. The training records list different initialization checkpoints and microbatch sizes (128 with KD, 64 without), although both runs use global batch 1024. We cannot verify that the runs used identical training examples, inference settings, or model-merging procedures. This comparison therefore does not isolate the effect of distillation. Multilingual results for the no-KD model and independent repetitions of the training runs are unavailable.

4.4 Inference throughput

Each throughput run used one NVIDIA H100 80 GB HBM3 GPU with BF16 precision and vLLM 0.24.0. Inputs are synthetic random token IDs of exact length, without padding. For each batch size and input length, we perform three warm-up calls followed by seven timed synchronous embedding calls. Throughput is BL divided by the median call duration, excluding model loading, tokenization, and input generation. Appendix C gives the software versions and engine settings.

The MoE configuration has the highest throughput among the five configurations in Table 4 at each evaluated length. Relative to dense 3B, its throughput is 1.28×, 1.25×, and 1.13× as high at 512, 1024, and 2048 tokens, respectively. All five configurations use vLLM. Because the backbones and runtime settings differ, these measurements cannot isolate the effect of sparse activation. The 480M model is not included in this throughput comparison.

4.5 External baseline comparison

Tables 5 and 6 compare Giga-Embeddings with public results from the MTEB Leaderboard. Baseline scores were obtained using different prompts and evaluation settings. Our 10B-A1.8B model has the highest Russian score in this comparison, while Qwen3-Embedding-8B has the highest English and multilingual scores and F2LLM-v2-14B the highest code score. Qwen3-Embedding-4B scores higher than our MoE model on English, multilingual, and code tasks, illustrating the quality trade-off associated with their throughput difference. The 480M encoder achieves a Russian MTEB score comparable to FRIDA (70.96 vs. 70.95) with approximately 42% fewer parameters, using their reported sizes

of 480M and 823M. The same checkpoint also supports English, multilingual, and code embedding tasks.

5 Conclusion

We present Giga-Embeddings, a family of dense and sparse bidirectional text encoders. The 10B-A1.8B model has the highest aggregate scores within the family on four MTEB suites and 13–28% higher measured batch-16 throughput than the dense 3B model. The MoE configuration is fastest among the five configurations in the throughput table, while its larger full checkpoint requires more weight storage than dense 3B. Compared with Qwen3-Embedding-4B, the MoE model has a higher Russian score and lower English, multilingual, and code scores. The 480M model achieves a Russian MTEB score comparable to FRIDA with approximately 42% fewer parameters and also supports multilingual embedding tasks. The family thus offers deployment options for different language, quality, and model-size requirements.

6 Limitations

Benchmark results are based on one run per model and do not provide uncertainty estimates for small score differences. The two 480M training variants differ in more than the distillation objective, and no repeated-seed study is available. The 480M multilingual score is recomputed from stored results produced with different MTEB versions; the exact model revision cannot be verified. Aggregate MTEB scores can also hide variation across tasks and languages.

The throughput benchmark uses synthetic token IDs and one GPU model, with seven timed repetitions per setting. It omits the 480M model and does not report latency or peak GPU memory. The models use different runtime settings, and the weight-storage estimates exclude activations and other runtime state.

The proprietary GigaChat pre-training checkpoints limit reproduction from raw data. The exact initialization checkpoints, teacher versions used to generate cached embeddings, and model-merging procedures are not fully documented. Training-time MoE load-balancing settings are also unavailable. We report no overlap between embedding-training and evaluation examples, but documentation of the checking procedure is unavailable. Future work should include matched training controls, evaluation datasets and model checkpoints with fixed version identifiers, and natural-text serving workloads.

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data inspire a versatile embedding model. In *Proceedings of the International Conference on Learning Representations (ICLR)*.

A Released Checkpoint Configurations

The public release configurations specify 28 layers, hidden size 1024, feed-forward size 3072, 16 attention heads, 8 key/value heads, and head dimension 64 for the 480M encoder. The 3B encoder has 36 layers, hidden size 2048, feed-forward size 11008, 16 attention heads, 8 key/value heads, and head dimension 128. Both use the Qwen3 bidirectional architecture family and a vocabulary of 128,256 tokens.

The MoE encoder uses the DeepSeek-V3 bidirectional architecture family with 26 layers, hidden size 1536, 32 attention heads, and the same vocabulary size. Its first layer has a dense feed-forward block of intermediate size 8960; the remaining 25 layers use 64 routed experts with intermediate size 1280, top-4 selection, and one shared expert. The released router configuration specifies sigmoid scores, `noaux_tc` selection, one expert group, normalization of selected routing weights, and a routing scale of 1.0. Multi-head latent attention uses a key/value low-rank dimension of 512, non-rotary and rotary query/key head dimensions of 128 and 64, and a value-head dimension of 192. These fields specify the released router configuration; they do not specify the training-time load-balancing objective.

The released safetensors metadata give total parameter counts of 483,717,632, 3,150,605,312, and 10,475,534,400 for the 480M, 3B, and MoE models, respectively. The MoE routed expert matrices contain

$$\begin{aligned} N_{\text{routed}} &= 25 \cdot 64 \cdot 3 \cdot 1536 \cdot 1280 \\ &= 9{,}437{,}184{,}000. \end{aligned} \tag{6}$$

weights, accounting for the gate, up, and down projections of each expert. The estimate includes all non-routed parameters, including shared experts and the embedding table, plus four of the 64 routed experts per token:

$$N_{\text{active}} = N_{\text{total}} - N_{\text{routed}} + \frac{4}{64} N_{\text{routed}} \tag{7}$$

$$= 1{,}628{,}174{,}400. \tag{8}$$

This is a parameter-counting convention, not a measure of FLOPs or GPU memory use.

Table 8: Training-example counts for selected 480M development runs. Consumed samples include repeated epochs; the rows do not describe a single final training mixture.

Development run	Prepared rows	Consumed
Giga pre-training	91,030,382	91,029,504
KaLM pre-training	299,034,744	299,024,384
F2LLM pre-training	10,597,526	10,597,376
Giga fine-tuning	1,131,959	1,131,520
Nemotron fine-tuning	3,355,596	3,354,624
KaLM fine-tuning	2,797,672	2,797,568
F2LLM fine-tuning	3,511,784	3,511,296
Multitask	2,439,873	7,317,504

B Training Data and Development Runs

Public datasets. The source recipes are Nemotron (Babakhin et al., 2025) and KaLM (Zhao et al., 2026). KaLM publishes a pre-training collection (HIT-TMG) and a fine-tuning collection (KaLM-Embedding). The Nemotron report describes a retrieval-pretraining mixture derived from Nemotron-CC-v2 and a separate fine-tuning mixture; its public fine-tuning release is Embed Nemotron Dataset V1. The training manifests list 25 KaLM pre-training files, 62 KaLM fine-tuning files, 14 Nemotron fine-tuning files, and 123 inputs for one multitask run. The 14 Nemotron entries are EmotionClassification, FEVER, GooAQ, HotpotQA, MAmmoTH2, MIRACL, MSMARCO, NaturalQuestions, NFCorpus, PAQ, SciFact, SQuAD, SyntheticClassificationData, and TriviaQA. These are file counts, not numbers of training examples. The examples used for embedding training do not overlap with the evaluation examples.

Development runs. Table 8 reports dataset sizes and consumed-sample counts for selected 480M development runs. Prepared rows are the dataset size after preparation; consumed samples count examples processed during training, including repeated epochs. The multitask counts come from a separate completed run, and the KaLM fine-tuning counts apply to no-KD and KaLM-12B-teacher runs. Completed Giga and F2LLM runs are also included. These counts cannot be assigned to the final released checkpoints because the complete model-merging history is unavailable.

Training settings. For the 14 documented compact-model runs, both the training and data-shuffling seeds were 42. Dataset shuffling was enabled, packing was disabled, and overlength exam-

ples were discarded. Maximum lengths were 512 tokens for pre-training and fine-tuning and 2048 for multitask training. The saved arguments do not specify per-source interleaving probabilities. The F2LLM pre-training run used global batch 1024; the Giga and KaLM pre-training runs used 16,384. Requested sample counts can include repeated examples and do not determine the source proportions after filtering. Language proportions, exact input dataset versions, and sampling weights for the final models are unavailable.

Distillation variants. The available initialization files for the two 480M variants differ in 308 of 310 tensors. The exact versions of these files used during training cannot be verified.

C Serving Benchmark Configuration

Each benchmark run used one H100 80 GB HBM3 GPU with BF16 precision, Python 3.12.13, PyTorch 2.11.0+cu129, CUDA runtime 12.9, Transformers 5.13.0, and vLLM 0.24.0. Runs were conducted in separate sessions using the same GPU model. The NVIDIA driver version and complete engine startup configuration are unavailable.

The benchmark generates token IDs on CPU with a fixed generator seed of 20260731. For each length L and batch size B, IDs are sampled uniformly from $[3, V)$; the first and last positions are set to IDs 1 and 2, without padding. The vocabulary bound is 128,000 for our models and Nemotron and 151,643 for Qwen3-Embedding-4B and F2LLM-v2-8B. These synthetic inputs do not represent natural text or the same texts encoded by different tokenizers. For timed repeat r, the seed is $20260731 + 100000r + L + B$; warm-up w uses $20260731 + w + L + B$.

Each (B, L) setting has three warm-ups followed by seven timed calls. Input generation precedes timing; a client-side CUDA synchronization precedes the wall-clock timer, which covers the synchronous embedding call and a subsequent synchronization. The embedding call includes engine scheduling and input/output handling, while loading, tokenization, and input generation are excluded. Throughput is $BL/\operatorname{median}(t_1, \ldots, t_7)$. We use mean pooling for our models and Nemotron, and last-token pooling for Qwen3 and F2LLM. All output embeddings are L2-normalized.

We use the pooling runner with embedding conversion and set `max_model_len=2048`, `max_num_seqs=16`, and `gpu_memory_utilization=0.90`. Prefix caching is disabled. Our dense encoders use bidirectional Qwen3 overrides. The MoE profile uses DeepSeek-V3 overrides, an encoder-only attention patch, eager execution, disabled FlashInfer autotuning, disabled MLA backend, and in-process execution; a DeepSeek rotary custom operation is disabled in favor of its native path. Values for other engine settings are unavailable. We do not attribute the throughput decline at 2048 tokens to a specific bottleneck. The timing logs do not include model-weight hashes, so the exact checkpoint versions cannot be verified.

D Russian Score Aggregation

The local and public 480M Russian aggregates use identical dataset revisions and task-level results on all 23 tasks. For MassiveIntentClassification, the test and validation scores are 0.826429 and 0.823463; for MassiveScenarioClassification they are 0.898184 and 0.892228. Selecting the test split when available gives a 23-task mean of 70.978526%. Averaging the validation and test scores of those two tasks gives 70.959130%, matching the public aggregate, a difference of 0.019396 percentage points. The difference between 70.98 and 70.96 therefore reflects split selection rather than changed task-level results. Both variants in Table 7 use the test-preferred aggregation.

E Multilingual Score Aggregation

The stored 480M results contain 46,933 subset/split scores across all 131 multilingual tasks. We use a fixed list of tasks, subsets, and splits, average the selected scores within each task, and then average the task means. This gives 56.799822%, displayed as 56.80†. The dataset revisions match the fixed task list. The results combine MTEB versions 2.12.30, 2.15.4, and 2.18.0 and do not identify a verifiable model revision. They therefore support recomputation of the aggregate but not verification of the exact checkpoint evaluated.

For LEMBPasskeyRetrieval, we average the eight splits named `test_256` through `test_32768`, obtaining 0.6625.

Applying the same task, subset, and split selection to the 3B and MoE results gives 63.890167% and 65.637226%, respectively, matching their public task-macro means. All 46,933 selected scores for each model match the public results.